## Scope and limitations of the study

- **User Account**
    - Each user can create a unique account using their own username and password.
- **Cloud Saves**
    - The save file of each user will be uploaded to the internet every time the game saves.
    - The save file of each user is accessible multi-platform.
    - Cloud saves prioritize the save with the highest amount of stars to prevent conflict.
- Offline Mode

  - Allows the player to play the game offline

Online Mode

- Allows the player to access the leaderboard and cloud saves.

- **Main Menu**
    - **Continue**
        - This allows the player to continue from their recently saved world.
    - Worlds
        - This will redirect the user to the saves menu to select their desired save or start a new save.
    - Settings
        - This will redirect the user to the settings panel to adjust the game's sound volume.
    - Quit

- This will close the game.

- **Settings Panel**
    - Volume Sliders
        - BGM Volume
            - Adjusts the volume for the background music of the game.
        - Effects Volume
            - Adjust the volume for the sound effects of the game.

    Fullscreen

    - Sets the game to full screen window.

    Credits

    - Lists the developers and individuals who are part of the game development along with references to the materials and software used.

- Worlds Menu
    - World Creation

        - Lets the player name the world before creation.

    - Delete
        - Deletes the chosen save file upon confirmation.

    * Rename

    - Allows the player to rename their worlds.

- **Save Slot**
        - Loads the chosen save file to continue its progress.

- Game Mechanics
- Procedural Map Generation
    - Single player maps are automatically created using algorithms instead of being manually designed which allows each environment to be unique.
- Missions
    - A list of gameplay objectives that gradually guides the player throughout the entire game.
    - Dungeon missions are separated into distinct levels of difficulties which encompasses a gradual flow of progression into learning programming.
    - Completing missions will give the player stars and the stars they have will determine their rank.
- Ranks
    - Ranking system will allow players to display their programming capabilities in the game.
    - Ranks can be increased by completing dungeons and earning blue stars.
- Completion Badges
    - Granted to the player upon completing extra dungeon missions. Only given once.

- Leaderboards
    - Sort available players by their rank.
    - Rankings are based on the blue stars the player has.
- Crafting
    - The user will be crafting various items that improve their current state or progress the missions provided.
    - The user can craft multiple categories of items which include materials, interactive objects, weapons, and foods. Some of which can have additional functionality using programming.
    - The users can craft items to build and progress.

Fishing

- Players can use fishing rods to catch fishes in bodies of water.

Farming

- Players can use tools to collect seeds and build farmland near water.

- Items
    - Materials
        - It can be used to craft materials and other craftable items.
    - Interactive Objects

* Crafting Stations

- Where the user can craft items.

* Furnace

- Allows the player to smelt ores and cook food using either wood or charcoal as fuel.

* Chest

- Allows the player to store items in a limited capacity.

* Color Crafter

- Allows the player to craft items with custom RGB colors.

- Displays the hex number for the selected block.

- Player can select between hex or RGB

Weapons

- Swords can be used to deal damage to enemies.

- Tools

  Tools such as the pickaxe, axe, and fishing rod can be used for specific purposes.

- Food

  - Can be used to heal health and restore stamina.

**Dungeons**

- Randomly generated areas that serve to give major progress for the user, challenging their skills in Lua or Python programming to unlock more mechanics.
- Rewards the player with blue stars upon completion.
- Serves as a guide to player's progress.
- Randomly spawns successively throughout the world and serve as the dungeon mechanic of the game. The next dungeon spawns after the last one are completed.
- Includes quizzes that can be solved openly depending on the user.

- Features Mini-Boss Battles in the last stage.

o Caves

Large Caves

- Generates in the overworld. Provides finite source of stone and ores

Pixie Caves

- Special caves with resources that regenerate after a period of time.

* Friendly Mobs

- Passive mobs like sheeps and wolves wander the overworld.

o Enemies

- Spawns randomly on the overworld and in dungeons.
- Enemies that spawn on dungeons do not adhere to the day and night cycle.

* Headless Fish

- Spawns in the overworld during the night and attacks by homing and dashing into the player.

* Desert Worm

- Spawns in the overworld regardless of daylight. Spawns and hides in the desert and attack the player when approached.

* Hitman

- Spawns in the dungeon. Fires projectiles and can charge straight into the player to deal damage.

* Slime

- Spawns in the overworld regardless of daylight. Bounces towards the player and

attacks them.

* Pixie

- Spawns in the Pixie Caves and attacks the player on sight.

- Day and Night Cycle
  - Determines enemy spawning rate and difficulty.
  - Enemies deal less damage during daytime and do normal damage at night.

- **Live Interpreter**
  - Serves as a built in IDE for the coding function in the game for both Algorithmica and Automatica.
  - Syntax Highlighting which will highlight the errors in programming syntax.
  - Interprets the code that is written by the player.

- Automatica
  - Robot-like craftables that allow the user to automate certain processes by programming it through Lua, through the Lua-focused dungeons.
  - Automaticas can be used to automate interaction with objects and aid the player in combat.
  - Automaticas required placed wires to interact with stations.
  - Automaticas have the option to be renamed.

- Algorithmica
  - Allows the user to automate actions in crafting stations, furnaces, and color crafters.

- LUA (Live Unit Application)
    - An advanced Automatica non-player character that introduces the user to the Lua language.
- Py-san
    - An advanced Algorithmica non-player character that introduces the user to the Python language.
- Autosave
    - The game will automatically save the player's progress periodically and during state changes such as entering caves and dungeons, as well as when the player quits the game.

Build Mode

- Can be toggled on or off to place down blocks.

- **Desktop and Mobile Application**
    - HUD (Heads-Up Display)
        - Pause
            - Allow the user to temporarily pause the game, showing the Pause Menu Panel.
        - Pause Menu Panel
            - Resume
                - Resumes the current game.
            - Settings

        - Shows the settings panel.

    - Lua Management

        - Allows the player to manage the LUA robots

    - Quit

        - Saves the game and directs the user to the Start Page.

- Mission Panel

    - Shows the user the current missions they can do to progress.

- Health Bar

    - Shows the user their current health.

Stamina Bar

- Shows the current stamina, which drains when running.

- Hotbar

    - Show the user their current held items.

**Inventory**

    - Shows the user their extra space and a smaller crafting area.

- Programming Encyclopedia

    - Contain details about programming concepts for Lua and Python learned through dungeons and custom functions for Algorithmica and Automatica.

- Code Editor Panel

    - A panel that prompts the user to code in Lua or Python to either progress in a dungeon or customize their Automatica and

Algorithmica.

- Crafting Panel
    - A panel for more advanced crafting, unlockable when the user crafts the crafting station.
- Map

  A representation of the world that shows the overworld, cave, and dungeons.
- On-screen Joystick
    - Allows movement for mobile users.

- Interact Button
    - A general button that changes depending on the items near the player's character.
- Build Mode Button

  - Can be toggled on or off if the player wants to build.